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ORIGINAL RESEARCH

Effectiveness and Safety of Early Rhythm Control in Patients With Atrial Fibrillation and Chronic Kidney Disease

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ABSTRACT

BACKGROUND Atrial fibrillation (AF) increases cardiovascular risk in patients with chronic kidney disease (CKD). The safety and efficacy of early rhythm control (ERC) in patients with CKD is not fully established.

OBJECTIVES This predefined secondary analysis of the EAST-AFNET 4 trial assessed the effectiveness and safety of ERC in patients with CKD defined by estimated glomerular filtration rate (GFR).

METHODS EAST-AFNET 4 randomized patients with recently diagnosed AF and comorbidities to ERC or usual care (UC). Key outcomes were analyzed by Kidney Disease Improving Global Outcomes defined CKD groups. The primary efficacy outcome combined cardiovascular death, stroke, hospitalization for worsening heart failure, or acute coronary syndrome. The safety outcome combined death, stroke, and serious rhythm control-related adverse events. Recurrent AF was a secondary outcome.

RESULTS Baseline creatinine was available in 2,742 of 2,789 (98.3%) patients. In this study, 23% had CKD (GFR: <60 mL/min/1.73 m^2). Patients with CKD were older (CKD: 74 $\pm$ 7.4 years; no CKD: 69 $\pm$ 8.3 years; $P < 0.001$), had higher CHA_2DS_2-VASc scores (CKD: 4 $\pm$ 1.4; no CKD: 3.2 $\pm$ 1.2; $P < 0.001$), and more primary outcome events over 5.1 years of follow-up (HR: 0.98 per mL GFR decrease [95% CI: 0.97-0.99 per mL GFR decrease]). ERC reduced the primary outcome with and without CKD (no CKD: ERC: 3.4%/100 patient-years; UC: 4.1%/100 patient-years; HR: 0.84; $P < 0.001$; CKD: ERC: 5.8%/100 patient-years; UC: 8.5%/100 patient-years; HR: 0.67; $P < 0.001$; $P_{\text{interaction}} = 0.133$). CKD increased safety outcomes without interaction with ERC ($P_{\text{interaction}} = 0.927$). Patients with CKD experienced more AF recurrences with UC ($P_{\text{interaction}} = 0.036$).

CONCLUSION ERC effectively and safely reduces cardiovascular events in patients with recently diagnosed AF and stroke risk factors with and without CKD. (Early Treatment of Atrial Fibrillation for Stroke Prevention Trial (EAST); NCT01288352) (JACC. 2026;88:227–241) © 2026 Published by Elsevier on behalf of the American College of Cardiology Foundation.

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ABBREVIATIONS AND ACRONYMS

AF = atrial fibrillation

CKD = chronic kidney disease

CKD-EPI = formula to calculate the estimated glomerular filtration rate (Chronic Kidney Disease Epidemiological Collaboration)

ECG = electrocardiogram

ERC = early rhythm control

GFR = glomerular filtration rate

KDIGO = Kidney Disease Improving Global Outcomes

SAE = serious adverse event

UC = usual care

Atrial fibrillation (AF) is a growing global epidemic.[1,2] AF increases the risk for cardiovascular events, including stroke, unplanned hospitalizations for heart failure, acute coronary syndrome, or cardiovascular death, and decreases quality of life.[3,4] Early rhythm control (ERC) therapy can reduce these events.[3] As a consequence, there has been a paradigm shift toward broader use of rhythm control in the management of patients with AF.[4-7]

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Chronic kidney disease (CKD) and AF aggravate each other.[4,6,8,9] Untreated AF can reduce kidney function,[10] whereas AF incidences rise stepwise with each CKD stage.[11] Shared comorbidities, elevated urea, electrolyte imbalance, and elevated concentrations of cardiovascular biomolecules also contribute to increased risk.[12-15] Consequently, the management of AF in patients with CKD includes important areas of uncertainty. In patients on hemodialysis, even data on the use of oral anticoagulation are insufficient for clear treatment recommendations.[16,17] Rate control without rhythm control therapy is often favored by clinicians in patients with moderate or severe chronic kidney disease or on dialysis.[18] Current practice guidelines do not provide specific treatment recommendations for patients with AF and CKD.[4,6]

To decrease therapeutic uncertainty with regard to rhythm control therapy, this prespecified secondary analysis of the EAST-AFNET 4 (Early Treatment of Atrial Fibrillation for Stroke Prevention Trial 4) trial determined the efficacy and safety of ERC in AF patients with and without CKD.

METHODS

TRIAL POPULATION AND INTERVENTION. The full methodology of the EAST-AFNET 4 trial has been published.[3] In this international, investigator-initiated, parallel-group, randomized, open-label trial with blinded outcome assessment, 2,789 patients were enrolled. Creatinine was quantified at baseline and at 2 years. Patients with end-stage kidney disease (Kidney Disease: Improving Global Outcomes [KDIGO] class V, glomerular filtration rate [GFR] <15 mL/min/1.73 m², or requiring dialysis) were not enrolled.[3] Eligible participants had an AF diagnosis within the preceding 12 months and presented with at least2 stroke risk factors, corresponding to a CHA_2DS_2-VASc score of ≥2. Patients were 1:1 randomized to ERC (n = 1,395) or usual care (UC) (n = 1,394). All analyses were performed by intention-to-treat. Rhythm control had to be initiated promptly after randomization in the ERC group, using antiarrhythmic drug therapy, cardioversion, or AF ablation. Patients allocated to UC were initially managed with rate control only. Rhythm control was possible if patients experienced AF-related symptoms that remained uncontrolled despite sufficient rate control. Cardiovascular comorbidities and therapies were assessed by site investigators at baseline and during prespecified follow-up visits.[3] Serum creatinine was collected at baseline. The EAST-AFNET 4 trial protocol was approved by ethical review boards (approval number: 2010-274-f-A) and

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appropriate authorities for all participating institutions. All participants gave written informed consent.

The first primary outcome was a composite of cardiovascular death, stroke (ischemic or hemorrhagic), or hospitalization for worsening heart failure or acute coronary syndrome, analyzed in a time-to-event analysis. A second primary outcome was defined as the number of nights spent in hospital per year. The primary safety outcome was a composite of death, stroke or serious adverse events (SAEs) attributed to rhythm control therapy. All SAEs were prospectively documented throughout the study and events were adjudicated centrally. Key secondary outcomes were each component of the combined primary outcome as well as heart rhythm, left ventricular function, quality of life, AF-related symptoms, and cognitive function, assessed with predefined and validated tests.[3] AF recurrences were documented by the study sites in the electronic case report form (eCRF). Recurrent AF was documented by site teams using clinically available methods including 12-lead electrocardiogram (ECG), Holter monitoring, and others. Patients randomized to ERC transmitted weekly telemetric ECGs using a patient-operated device to a core laboratory (Vitatron, Munich, Germany). Sites were asked to perform a triggered visit when these ECGs showed AF.[3] All patients had a 12-lead ECG at 1 and 2 years of follow-up.[19,20]

CKD DEFINITIONS. For this analysis, which was prespecified in the statistical analysis plan[3] and contains exploratory elements, all patients were stratified into KDIGO-defined CKD classes using baseline estimated GFR, determined by the 2009 CKD-EPI equation based on serum creatinine.[21] CKD was divided into 5 categories (KDIGO class I, II, IIIa, IIIb, and IV[22]) by estimated GFR. Effects of ERC and UC between the randomized groups were compared by presence of CKD in a dichotomous fashion, by comparing KDIGO kidney disease classes, and by integrating estimated GFR as a continuous variable. For simplicity, KDIGO classes I to II (GFR $\geq$90 mL/min/1.73 m^2 and 60-89 mL/min/1.73 m^2) will be called no CKD, and KDIGO classes III to V (IIIa: 45-59 mL/min/1.73 m^2; IIIb: 30-44 mL/min/1.73 m^2; IV: $<$30 mL/min/1.73 m^2) will be called CKD. Patients on chronic dialysis were not included. There were no patients with history of kidney transplantation in this trial.

OUTCOMES AND STATISTICAL METHODS. All patients randomized in EAST-AFNET4 were categorized on the basis of serum creatinine transformed to GFR with the 2009 CKD-EPI-Formula using R package *nephron*.[23] Baseline characteristics of patients are summarized using descriptive statistical methods and tested for difference between no CKD and CKD strata. Continuous variables were described by mean and SD or by median with 1st and 3rd quantiles. Categorical data were summarized as absolute and relative frequencies. *P* values were obtained from analysis of variance tests based on mixed linear or logistic regression models with center identification (ID) as random effect. Nominal variables were tested with Pearson chi-square test.

The first primary outcome of the EAST-AFNET4 trial as well as its components of time-to-event outcomes, that is, time to cardiovascular death, time to first stroke, time to first hospitalization for worsening heart failure, and time to first hospitalization for acute coronary syndrome, were analyzed with a Cox proportional hazards model with a frailty term for the cluster center. The treatment effects are expressed as cause-specific HRs and corresponding 95% CIs.

The second primary outcome, nights in the hospital, was analyzed using a negative binomial mixed model including the log of follow-up time as the offset and site as random effect. The treatment effect is shown as incidence rate ratio and 95% CI. Following practice in all EAST-AFNET 4 analyses, "study center" was included as a random effect to account for potential heterogeneity across centers, including differences in rhythm control practice patterns, regional variations in healthcare delivery, and differences in patient management, outcome ascertainment, or reporting.[24]

For continuous and categorical secondary outcomes (change from baseline to 24 months) including left ventricular ejection fraction, European Quality of Life 5 Dimensions score, 12-item Short Form Survey (mental and physical score), Montreal Cognitive Assessment score, sinus rhythm and symptoms, baseline adjusted mixed linear or logistic models were used after applying a multivariable imputation with chained equations algorithm with 60 imputations of missing values for a prespecified set of variables on the basis of suggestions by White, Royston, and Wood (see statistical analyses plan in the supplement of Kirchhof et al[3]) and expressed as the adjusted mean difference and 95% CI or the OR and 95% CI. Center was included as random effect.

Further outcomes that were included are time-to-event analyses for the primary outcome of the

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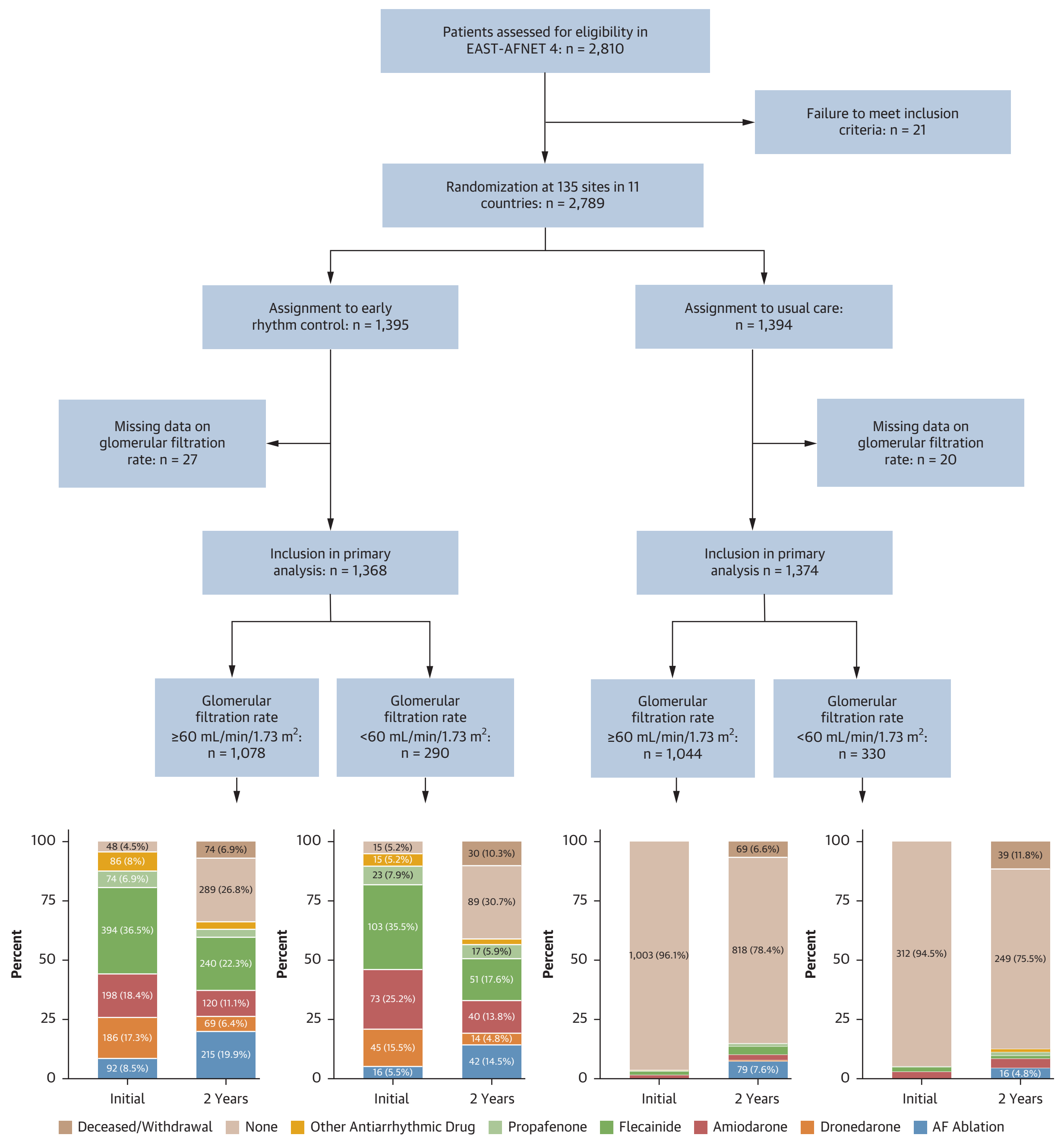
FIGURE 1 **CONSORT Diagram of the Analysis (Including Screening, Randomization, Prespecified Analysis in Respect to GFR and Treatment)**



AF = atrial fibrillation; GFR = glomerular filtration rate.

CASTLE-AF (Catheter Ablation versus Standard Conventional Therapy in Patients with Left Ventricular Dysfunction and Atrial Fibrillation) trial (death or heart failure hospitalization), CABANA (Catheter Ablation vs Antiarrhythmic Drug Therapy for Atrial Fibrillation) trial (death, disabling stroke, serious bleeding, or cardiac arrest), and time-to-event analysis of recurrent AF as well as sinus rhythm at 12 months. These outcomes were analyzed as described already in this article.

TABLE 1 Demographic and Clinical Characteristics at Baseline

	Overall (N = 2,742)	No Kidney Disease, GFR ≥60 mL/min/1.73 m² (n = 2,122 [77%])	Kidney Disease, GFR <60 mL/min/1.73 m² (n = 620 [23%])	*P* Value
Age, y	70 ± 8.3	69 ± 8.3	74 ± 7.3	<0.001
Sex				<0.001
Female	1,271/2,742 (46%)	932/2,122 (44%)	339/620 (55%)	
Male	1,471/2,742 (54%)	1,190/2,122 (56%)	281/620 (45%)	
Body mass index, calculated; kg/m²	29.3 ± 5.4	29.3 ± 5.4	29.3 ± 5.3	0.52
AF type				0.83
First episode	1,035/2,742 (38%)	795/2,122 (37%)	240/620 (39%)	
Paroxysmal	971/2,742 (35%)	757/2,122 (36%)	214/620 (35%)	
Persistent or long-standing persistent	736/2,742 (27%)	570/2,122 (27%)	166/620 (27%)	
Sinus rhythm at baseline	1,484/2,741 (54%)	1,163/2,122 (55%)	321/619 (52%)	0.13
Median days since AF diagnosis	35.0 (6.0-112.0)	36.0 (6.0-111.0)	31.0 (5.5-112.0)	<0.001
Absence of AF symptoms	794/2,597 (31%)	606/2,003 (30%)	188/594 (32%)	0.73
Previous pharmacologic or electrical cardioversion	1,074/2,711 (40%)	806/2,099 (38%)	268/612 (44%)	0.12
Concomitant cardiovascular conditions				
Previous stroke or transient ischemic attack	322/2,742 (12%)	244/2,122 (11%)	78/620 (13%)	0.39
At least mild cognitive impairment	1,149/2,632 (44%)	854/2,037 (42%)	295/595 (50%)	<0.001
Arterial hypertension	2,412/2,742 (88%)	1,846/2,122 (87%)	566/620 (91%)	0.004
Systolic blood pressure, mm Hg	137 ± 19.2	138 ± 19.3	134 ± 18.6	<0.001
Diastolic blood pressure, mm Hg	81 ± 12.0	82 ± 12.0	78 ± 11.7	<0.001
CHA_2DS_2-VASc score	3.4 ± 1.3	3.2 ± 1.2	4.0 ± 1.4	<0.001
Chronic kidney disease of KDIGO stage 3 or 4	350/2,742 (13%)	7/2,122 (0.3%)	343/620 (55%)	<0.001
Diabetes mellitus	684/2,742 (25%)	495/2,122 (23%)	189/620 (30%)	<0.001
Left ventricular ejection fraction	58.8 ± 10.0	59.0 ± 9.8	58.1 ± 10.9	0.004
Coronary artery disease	473/2,742 (17%)	328/2,122 (15%)	145/620 (23%)	<0.001
Peripheral artery disease	118/2,742 (4.3%)	69/2,122 (3.3%)	49/620 (7.9%)	<0.001
Valvular heart disease	1,239/2,740 (45%)	915/2,120 (43%)	324/620 (52%)	0.002
Myocardial infarction	251/2,741 (9.2%)	174/2,121 (8.2%)	77/620 (12%)	0.002
Stable HF	787/2,742 (29%)	561/2,122 (26%)	226/620 (36%)	<0.001
HFrEF	131/787 (17%)	88/561 (16%)	43/226 (19%)	0.23
HFmrEF	208/787 (56%)	160/561 (29%)	48/226 (21%)	0.014
HFpEF	438/787 (56%)	307/561 (55%)	131/226 (58%)	0.32
Medication at discharge				
Oral anticoagulation with DOAC or VKA	2,480/2,742 (90%)	1,925/2,122 (91%)	555/620 (90%)	0.37
VKA	939/2,742 (34%)	727/2,122 (34%)	212/620 (34%)	0.085
Apixaban	386/2,742 (14%)	295/2,122 (14%)	91/620 (15%)	0.74
Edoxaban	50/2,742 (1.8%)	41/2,122 (1.9%)	9/620 (1.5%)	0.55
Dabigatran	324/2,742 (12%)	255/2,122 (12%)	69/620 (29%)	0.50
Rivaroxaban	793/2,742 (29%)	614/2,122 (29%)	179/620 (29%)	0.15
Digoxin or digitoxin	129/2,742 (4.7%)	94/2,122 (4.4%)	35/620 (5.6%)	0.15
Beta-blockers	2,221/2,742 (81%)	1,701/2,122 (80%)	520/620 (84%)	0.20
ACE inhibitors or angiotensin II receptor blocker	1,905/2,742 (69%)	1,433/2,122 (68%)	472/620 (76%)	<0.001
Mineralocorticoid receptor antagonist	182/2,742 (6.6%)	112/2,122 (5.3%)	70/620 (11%)	<0.001
Diuretic	1,108/2,742 (40%)	763/2,122 (36%)	345/620 (56%)	<0.001
Statin	1,182/2,742 (43%)	891/2,122 (42%)	291/620 (47%)	0.028
Platelet inhibitor	453/2,742 (17%)	325/2,122 (15%)	128/620 (21%)	<0.001
SGLT2i	1/2,742 (<0.1%)	0/2,122 (0%)	1/620 (0.2%)	
GLP1	1/2,742 (<0.1%)	1/2,122 (<0.1%)	0/620 (0%)	
Planned therapy for rhythm control at baseline				0.072
AAD	1,254/2,742 (46%)	977/2,122 (46%)	277/620 (45%)	
Ablation	110/2,742 (4.0%)	94/2,122 (4.4%)	16/620 (2.6%)	
None	1,378/2,742 (50%)	1,051/2,122 (50%)	327/620 (53%)	

Values are mean ± SD, n/N (%), or median (IQR).

AAD = antiarrhythmic drugs; ACE = angiotensin-converting enzyme; AF = atrial fibrillation; DOAC = direct oral anticoagulant; GFR = glomerular filtration rate; HF = heart failure; HFmrEF = heart failure with mildly reduced ejection fraction; HFpEF = heart failure with preserved ejection fraction; HFrEF = heart failure with reduced ejection fraction; KDIGO = Kidney Disease Improving Global Outcomes; VKA = vitamin K antagonist.

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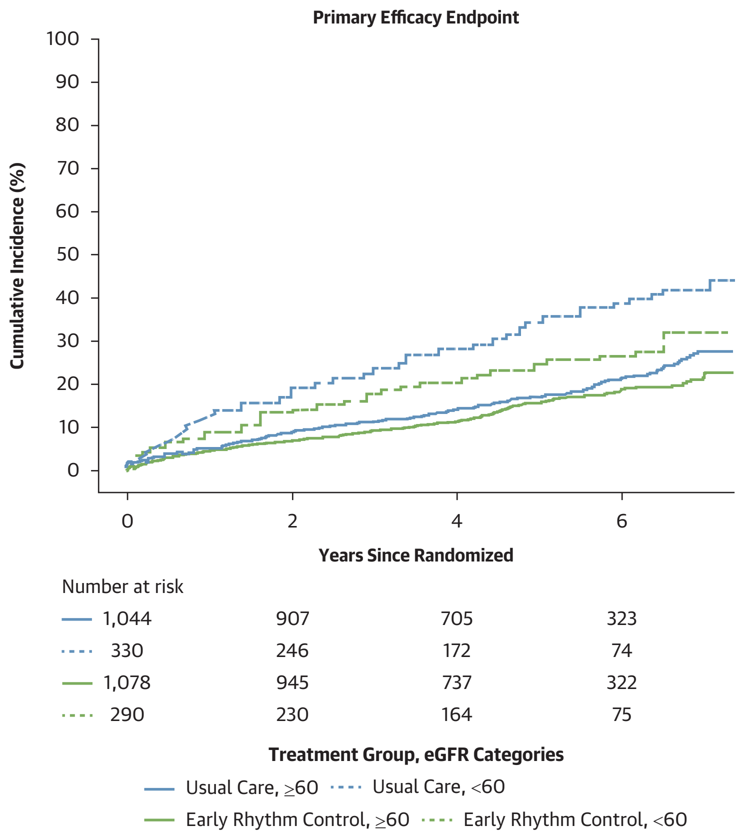
FIGURE 2 **Aalen-Johansen Curve Showing Time to a First Primary Outcome, a Composite of Cardiovascular Death, Stroke, or Unplanned Hospitalization for Heart Failure or Acute Coronary Syndrome, by Randomized Group and Kidney Disease**



eGFR = estimated glomerular filtration rate.

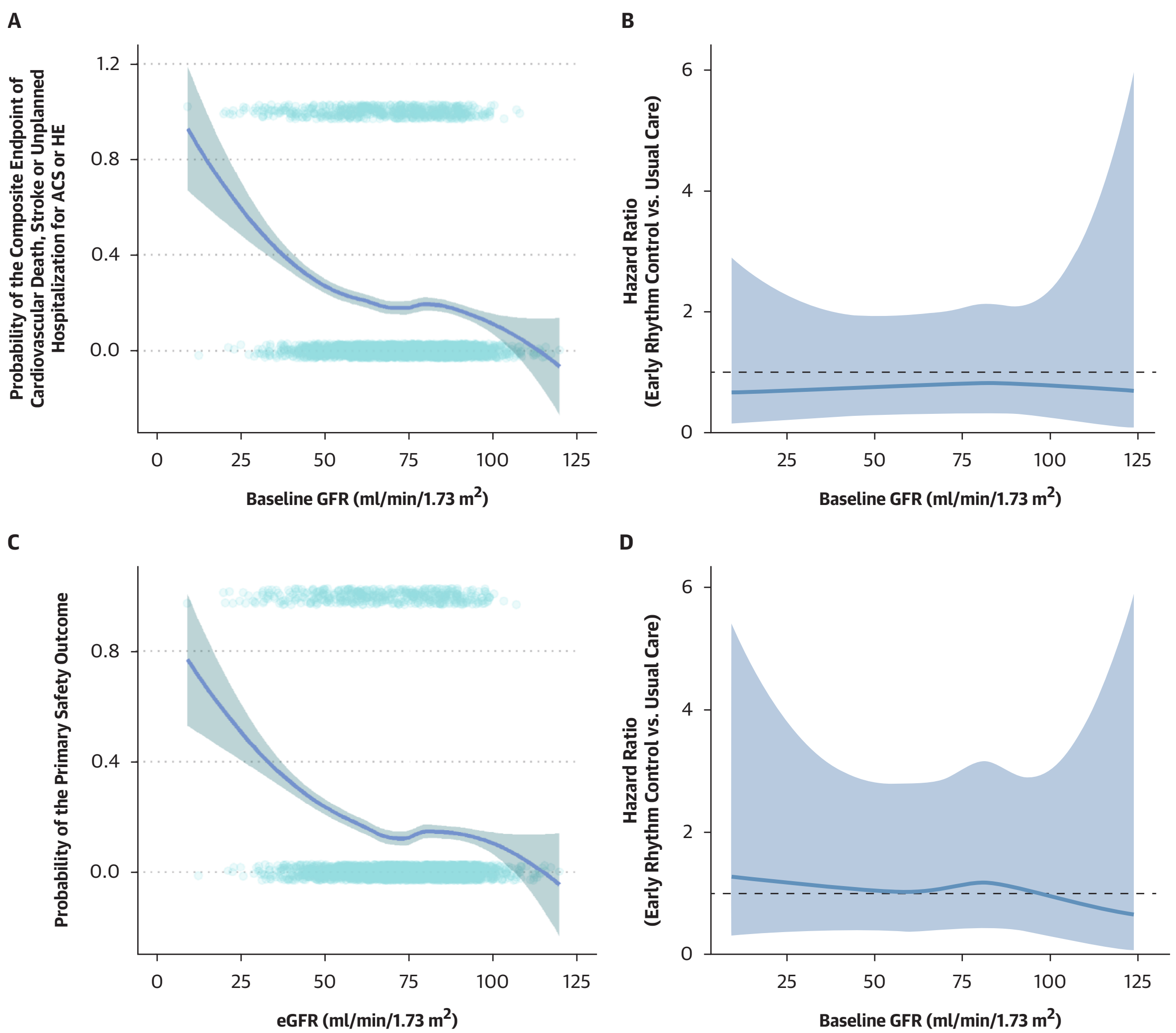
TABLE 2 **Number of Patients and Treatment Effects With a First Primary Efficacy Outcome, a Composite of CV Death, Stroke, or Unplanned Hospitalization by Randomized Group and by Presence or Absence of CKD**

	GFR ≥60 mL/min/1.73 m²			GFR <60 mL/min/1.73 m²			
	ERC	UC	Treatment Effect	ERC	UC	Treatment Effect	*P* Value Interaction
First primary outcome - events/person-y (incidence/100 person-y)	173/5,068 (3.4)	204/4,949 (4.1)	0.84 (0.68, 1.02)	72/1,238 (5.8)	110/1,293 (8.5)	0.67 (0.49, 0.9)	0.133
Components of first primary outcome - events/person-y (incidence/100 person-y)							
Death from CV causes	43/5,429 (0.8)	58/5,352 (1.1)	0.74 (0.5, 1.1)	22/1,386 (1.6)	35/1,545 (2.3)	0.67 (0.39, 1.15)	0.564
Stroke	31/5,347 (0.6)	42/5,268 (0.8)	0.74 (0.46, 1.17)	9/1,366 (0.7)	20/1,497 (1.3)	0.49 (0.22, 1.07)	0.291
Hospitalization with worsening of HF	94/5,232 (1.8)	101/5,118 (2)	0.91 (0.69, 1.21)	43/1,295 (3.3)	67/1,349 (5)	0.65 (0.44, 0.96)	0.118
Hospitalization with ACS	39/5,324 (0.7)	48/5,226 (0.9)	0.8 (0.52, 1.22)	13/1,338 (1)	17/1,499 (1.1)	0.87 (0.42, 1.8)	0.725
Second primary outcome - nights spent in hospital/y	5.26 ± 21.5	4.1 ± 11.9	1.09 (0.94, 1.26)	8.17 ± 24.4	8.35 ± 23.7	1.11 (0.85, 1.44)	0.923

Values are n/N (%) or mean ± SD. The second primary outcome, nights spent in hospital, is shown in the last line.

ACS = acute coronary syndrome; CKD = chronic kidney disease; CV = cardiovascular; ERC = early rhythm control; UC = usual care; other abbreviations as in Table 1.

FIGURE 3 **Analysis of the Treatment Effect Across the Spectrum of Kidney Functions**



(A) LOESS analysis illustrating the probability of the primary efficacy endpoint in relation to the eGFR. (B) HR (primary efficacy endpoint, ERC vs UC) across the range of baseline eGFR estimated from a Cox proportional hazards model including a restricted cubic spline with 4 knots and a shared frailty term for center. Shaded area represents 95% CIs. (C) LOESS analysis illustrating the probability of the primary safety outcome in relation to the eGFR. (D) HR (primary safety outcome, ERC vs UC) across the range of baseline eGFR estimated from a Cox proportional hazards model including a restricted cubic spline with 4 knots and a shared frailty term for center. Shaded area represents 95% CIs. ACS = acute coronary syndrome; ERC = early rhythm control; eGFR = estimated glomerular filtration rate; GFR = estimated glomerular filtration rate; HF = heart failure; LOESS = locally estimated scatterplot smoothing; UC = usual care.

Safety outcomes were analyzed through mixed logistic models with center ID as random effect.

RESULTS

BASELINE CHARACTERISTICS. Baseline GFR was available in 2,742 of 2,789 patients (98%), including 1,368 of 2,742 assigned to ERC and 1,374 of 2,742 assigned to usual care (Figure 1). CKD was present in 620 patients (23%). Patients with CKD were older, included more women, had higher rates of comorbidities as assessed by the CHA_2DS_2-VASc-Score, exhibited more cognitive impairment, and had a shorter time since AF diagnosis (Table 1). Treatments included guideline-recommended therapy of concomitant conditions and anticoagulation in most patients. The use of anticoagulation did not differ between patients with and without CKD.

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TABLE 3 First Primary, Second Primary, and Key Secondary Outcomes in all 4 Analyzed Kidney Function Classes

	GFR ≥90 mL/min/1.73 m²		GFR 89-60 mL/min/1.73 m²		GFR 59-45 mL/min/1.73 m²		GFR <45 mL/min/1.73 m²		*P* Value Interaction
	ERC	UC	ERC	UC	ERC	UC	ERC	UC	
First primary outcome	26/1,118 (2.3)	32/994 (3.2)	147/3,950 (3.7)	172/3,955 (4.3)	43/944 (4.6)	67/977 (6.9)	29/294 (9.9)	43/317 (13.6)	0.485
Components of first primary outcome - events/person-y (incidence/100 person-y)									
Death from CV causes	10/1,153 (0.9)	10/1,051 (1)	33/4,276 (0.8)	48/4,301 (1.1)	9/1,052 (0.9)	16/1,145 (1.4)	13/335 (3.9)	19/400 (4.8)	0.399
Stroke	5/1,147 (0.4)	7/1,037 (0.7)	26/4,201 (0.6)	35/4,230 (0.8)	4/1,038 (0.4)	16/1,104 (1.4)	5/327 (1.5)	4/393 (1)	0.143
Hospitalization with worsening of HF	9/1,136 (0.8)	15/1,021 (1.5)	85/4,096 (2.1)	86/4,098 (2.1)	26/986 (2.6)	40/1,014 (3.9)	17/309 (5.5)	27/335 (8.1)	0.193
Hospitalization with ACS	8/1,138 (0.7)	7/1,031 (0.7)	31/4,186 (0.7)	41/4,195 (1)	10/1,014 (1)	6/1,126 (0.5)	3/324 (0.9)	11/373 (2.9)	0.141
Nights spend in hospital/y	8.08 ± 36.9	3.08 ± 8.5	4.52 ± 15	4.36 ± 12.6	5.95 ± 15.5	8.27 ± 26.2	14.6 ± 39.8	8.57 ± 16.1	**0.002**
Key secondary outcomes at 2 y									
Change in LVEF	2.73 ± 9.8	0.73 ± 9.2	1.35 ± 9.7	0.68 ± 9.9	0.72 ± 10.3	1.33 ± 10.4	1.26 ± 8.4	0.05 ± 9.3	0.445
Change in EQ-5D score	−2.32 ± 21.2	−0.3 ± 19.4	−0.42 ± 21.1	−2.81 ± 21.6	−0.67 ± 20.7	−5.04 ± 25.9	−4 ± 27.9	−0.19 ± 24.4	0.594
Change in SF-12 Mental Score	1.42 ± 11.4	2.2 ± 10	0.69 ± 10.2	1.57 ± 9.7	−0.43 ± 10.3	1.33 ± 10.5	0.87 ± 14.1	1.99 ± 13.3	0.558
Change in SF-12 Physical Score	0.21 ± 8.8	0.8 ± 9.1	0.16 ± 8.5	−0.18 ± 7.9	1.39 ± 8.4	0.99 ± 8.4	0.13 ± 8.6	−2.28 ± 7.9	0.288
Change in MoCA score	0.33 ± 3.2	0.05 ± 3.2	0.01 ± 3.2	0.08 ± 3.1	0.25 ± 3.8	0.28 ± 3.7	0.07 ± 2.8	−0.15 ± 3.1	0.898
Sinus rhythm	160/182 (87.9)	121/176 (68.8)	573/705 (81.3)	414/688 (60.2)	135/169 (79.9)	111/188 (59)	41/51 (80.4)	33/68 (48.5)	0.803
Asymptomatic	144/187 (77)	130/184 (70.7)	545/730 (74.7)	526/710 (74.1)	123/174 (70.7)	135/192 (70.3)	38/52 (73.1)	49/70 (70)	0.709
Other									
Sinus rhythm at 12 mo	170/190 (89.5)	137/185 (74.1)	623/745 (83.6)	478/735 (65)	154/180 (85.6)	125/196 (63.8)	48/56 (85.7)	40/72 (55.6)	0.543
AF recurrence	80/760 (10.5)	81/687 (11.8)	336/2,759 (12.2)	413/2,453 (16.8)	86/678 (12.7)	114/663 (17.2)	23/240 (9.6)	51/202 (25.3)	**0.031**

Values are or n/N (%) or mean ± SD. *P* values in **bold** are significant. Secondary outcomes are exploratory and therefore reported without formal statistical testing.
LVEF = left ventricular ejection fraction; other abbreviations are as in Tables 1 and 2.

PRIMARY EFFICACY OUTCOMES. ERC therapy was effective in reducing the first primary outcome with and without CKD ($P_{\text{interaction}} = 0.133$) (Figure 2). The result was consistent across the components of the primary outcome (Table 2). The treatment effect appeared consistent across the spectrum of kidney functions when analyzed in a continuous fashion (Figures 3A and 3B). The effectiveness of ERC was maintained across all 4 CKD classes (Table 3) ($P_{\text{interaction}} = 0.485$) and in a sensitivity analysis splitting patients into 3 GFR groups (Supplemental Table 1). Results were consistent when the primary outcome of the analysis was replaced by the primary outcomes of the CASTLE-AF[25] or CABANA[26] trials (Table 3). Patients with CKD had a higher risk of cardiovascular events and spent more days in hospital than patients without CKD (Figure 2, Tables 2 and 3).

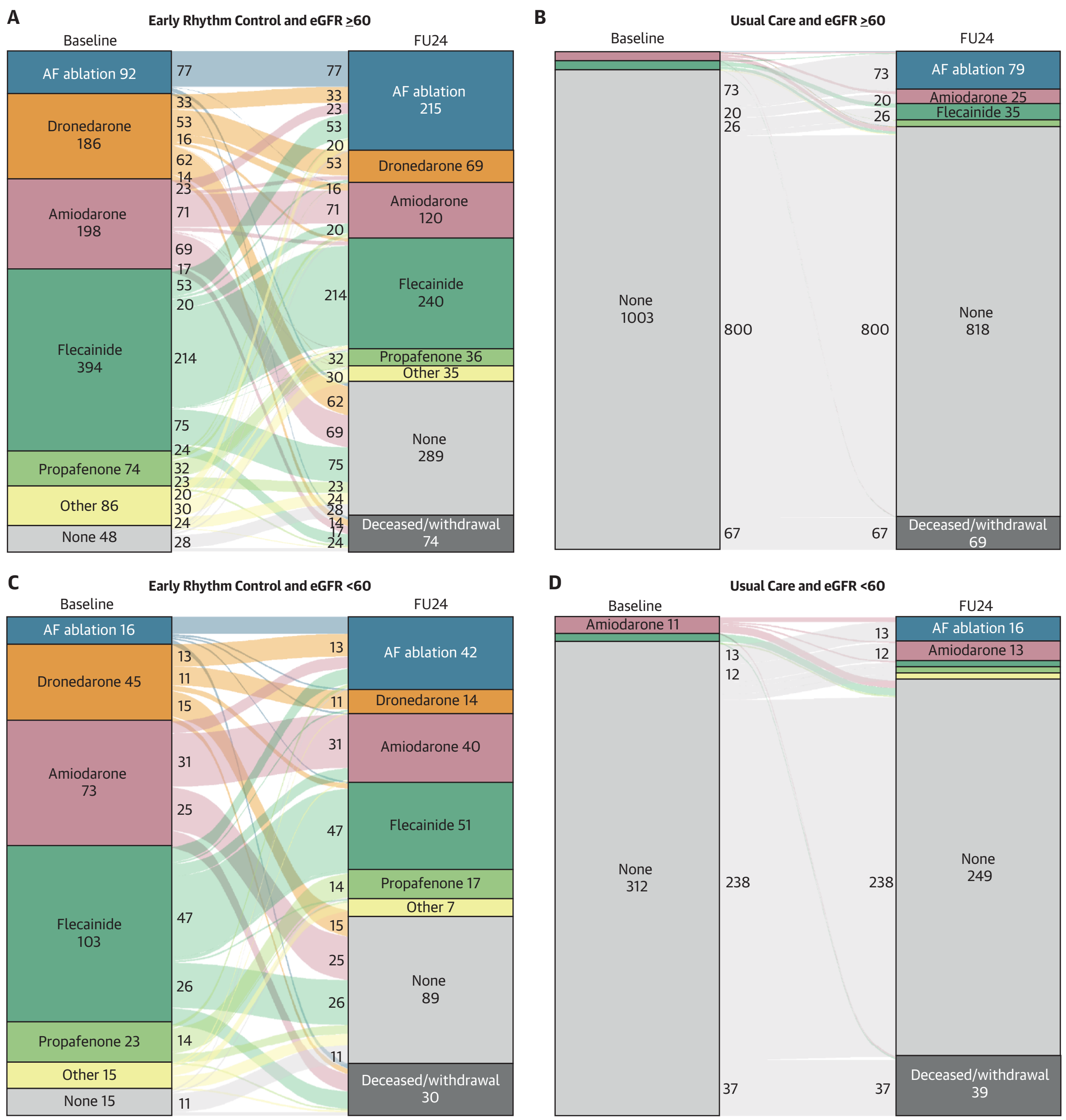
SECONDARY OUTCOMES. ERC was more effective in preventing recurrent AF episodes in patients with CKD (no CKD: HR: 0.82; 95% CI: 0.72; 0.94; CKD: HR: 0.65; 95% CI: 0.51; 0.83; $P_{\text{interaction}} = 0.036$). This was driven by more AF recurrences with UC in patients with CKD (CKD: ERC: 109 of 918 [11.8%]; AF recurrences/patient-year; UC: 165 of 865 [19.1%]; AF recurrences/patient-year; no CKD: ERC: 416 of 3,519 [11.8%]; AF recurrences/patient-year; UC: 494 of 3,140 [15.7%]; AF recurrences/patient-year). AF burden on ERC was not affected by CKD (HR: 0.92; 95% CI: 0.66; 1.28) (Table 4). Heart failure (HR: 1.50; 95% CI: 1.10; 2.04; $P = 0.011$) and absence of sinus rhythm at baseline predicted AF burden (HR: 4.20; 95% CI: 3.17; 5.57; $P < 0.001$).

Other key secondary outcomes were not different between ERC and UC arms, regardless of whether patients had CKD or not (Supplemental Tables 2 and 3). Neither change in left ventricular ejection fraction, nor change in physical, mental, or quality of life scores were relevantly different (Table 3).

SAFETY OUTCOMES. CKD increased safety events (death, stroke, or SAEs attributed to rhythm control therapy) (Table 5, Figure 3C and 3D). ERC maintained its safety with and without CKD ($P_{\text{interaction}} = 0.927$). The results were consistent for each component of the safety outcome (Table 5, Supplemental Table 3).

TYPES OF RHYTHM CONTROL PER GFR GROUP. Amiodarone was more often used in patients with CKD (CKD: 73 of 290 patients [25%]; no CKD: 198 of 1,078 patients [18%]). Dronedarone, flecainide, and propafenone were equally used across CKD groups. AF ablation was less commonly performed in patients with CKD (CKD: 5.5%; no CKD: 8.5%). At 2 years of follow-up, AF ablation continued to be less often used in patients with CKD (42 of 260 [16%] patients) than in patients without CKD (215 of 1,004 [21%]

FIGURE 4 **Sankey Plot illustrating Distribution of Antiarrhythmic Treatment in the ERC and UC Group Stratified After GFR (Baseline to 2 Years of Follow-Up)**



FU24 = 2 years of follow-up; other abbreviations as in Figures 1, 2, and 3.

patients). Antiarrhythmic drug use decreased in all patients, regardless of kidney disease status (Figure 4).

CHANGE IN KIDNEY FUNCTION AT 2 YEARS AND INTERACTION WITH ERC. The change in GFR from baseline to the 2-year follow-up (Table 6) was a median of −3 mL/min/1.73 m^2 (95% CI: −13 to 3 mL/min/1.73 m^2) with ERC and −2 mL/min/1.73 m^2 (−10 to 4 mL/min/1.73 m^2) with UC in patients with GFR ≥60 mL/min/1.73 m^2. In patient with reduced kidney function (GFR: <60 mL/min/1.73 m^2), GFR at 2 years changed a median of 0 mL/min/1.73 m^2 (−7 to 7 mL/min/1.73 m^2) with ERC and 1 mL/min/1.73 m^2 (−4 to 10 mL/min/1.73 m^2) with UC (Figure 5). There

TABLE 4 AF Burden of Patients in the ERC Group Presented as Percentage of Telemetric, Patient-Operated ECGs in AF During Follow-Up and Divided in Quartiles of AF Percentages

	Overall (N = 1,178)	Quartile 1 [0-0.936] (n = 295 [25%])	Quartile 2 [0.936-6.56] (n = 294 [25%])	Quartile 3 [6.56-21.7] (n = 294 [25%])	Quartile 4 [21.7-100] (n = 295 [25%])
GFR					
Mean ± SD	73 ± 16	74 ± 17	74 ± 16	73 ± 17	72 ± 16
Median (Q1-Q3)	74 (62-86)	76 (63-87)	74 (63-86)	73 (62-87)	72 (61-85)
GFR groups (binary)					
≥60 mL/min/1.73 m^2	925/1,163 (80%)	234/294 (80%)	229/288 (80%)	237/290 (82%)	225/291 (77%)
<60 mL/min/1.73 m^2	238/1,163 (20%)	60/294 (20%)	59/288 (20%)	53/290 (18%)	66/291 (23%)
GFR groups (3 groups)					
≥60 mL/min/1.73 m^2	925/1,163 (80%)	234/294 (80%)	229/288 (80%)	237/290 (82%)	225/291 (77%)
45-59 mL/min/1.73 m^2	179/1,163 (15%)	45/294 (15%)	45/288 (16%)	36/290 (12%)	53/291 (18%)
<45 mL/min/1.73 m^2	59/1,163 (5.1%)	15/294 (5.1%)	14/288 (4.9%)	17/290 (5.9%)	13/291 (4.5%)
Age					
Mean ± SD	70 ± 8	69 ± 9	70 ± 8	70 ± 8	71 ± 9
Median (Q1-Q3)	71 (65-76)	70 (63-75)	71 (65-75)	70 (66-75)	72 (66-77)
Sex					
Female	533/1,178 (45%)	147/295 (50%)	138/294 (47%)	129/294 (44%)	119/295 (40%)
Male	645/1,178 (55%)	148/295 (50%)	156/294 (53%)	165/294 (56%)	176/295 (60%)
Sinus rhythm at baseline	656/1,178 (56%)	230/295 (78%)	199/294 (68%)	128/294 (44%)	99/295 (34%)
Stable HF	325/1,178 (28%)	72/295 (24%)	65/294 (22%)	93/294 (32%)	95/295 (32%)

Values n/N (%), mean ± SD, or median (IQR).
Abbreviations as in Tables 1 and 2.

TABLE 5 Primary Safety Events and SAEs Divided Into Both Kidney Function and Treatment Groups

	GFR ≥60 mL/min/1.73 m^2		GFR <60 mL/min/1.73 m^2		*P* Value Interaction	*P* Value Without Interaction
	ERC (n = 1,078)	UC (n = 1,044)	ERC (n = 290)	UC (n = 330)		
Primary composite safety outcome	149 (13.8%)	139 (13.3%)	79 (27.2%)	83 (25.2%)	0.927	<0.001
Stroke	31 (2.9%)	42 (4.0%)	9 (3.1%)	20 (6.1%)	0.439	0.146
Death	82 (7.6%)	102 (9.8%)	54 (18.6%)	61 (18.5%)	0.332	<0.001
SAE related to rhythm control therapy	46 (4.3%)	9 (0.9%)	21 (7.2%)	10 (3.0%)	0.166	0.002
SAE related to antiarrhythmic drug therapy						
Nonfatal cardiac arrest	0 (0.0%)	0 (0.0%)	1 (0.3%)	1 (0.3%)		
Antiarrhythmic Drug toxicity	7 (0.6%)	1 (0.1%)	2 (0.7%)	2 (0.6%)	0.222	0.344
Drug-induced bradycardia	11 (1.0%)	2 (0.2%)	3 (1.0%)	3 (0.9%)	0.168	0.313
Atrioventricular block	1 (0.1%)	0 (0.0%)	1 (0.3%)	0 (0.0%)		
Torsade-de-pointes tachycardia	0 (0.0%)	0 (0.0%)	1 (0.3%)	0 (0.0%)		
SAE related to AF ablation						
Pericardial tamponade	2 (0.2%)	0 (0.0%)	1 (0.3%)	0 (0.0%)		
Major bleeding related to AF ablation	1 (0.1%)	0 (0.0%)	5 (1.7%)	0 (0.0%)		
Nonmajor bleeding related to AF ablation	1 (0.1%)	2 (0.2%)	0 (0.0%)	0 (0.0%)		
SAE of special interest related to RC therapy						
Blood pressure-related event	1 (0.1%)	0 (0.0%)	0 (0.0%)	0 (0.0%)		
Hospitalization for AF	8 (0.7%)	0 (0.0%)	3 (1.0%)	3 (0.9%)	0.989	0.102
Other CV event	1 (0.1%)	0 (0.0%)	4 (1.4%)	1 (0.3%)		
Other event	0 (0.0%)	1 (0.1%)	1 (0.3%)	2 (0.6%)		
Syncope	4 (0.4%)	1 (0.1%)	0 (0.0%)	0 (0.0%)		
Hospitalization for worsening of HF	3 (0.3%)	0 (0.0%)	0 (0.0%)	0 (0.0%)		
Implantation of a pacemaker/defibrillator	7 (0.6%)	3 (0.3%)	1 (0.3%)	1 (0.3%)	0.842	<0.001

Values are mean ± SD or n (%).
RC = rhythm control; SAE = serious adverse event; other abbreviations as in Tables 1 and 2.

was no relevant impact of kidney function on the longitudinal GFR change (−1.06 mL/min/1.73 m^2; 95% CI: −2.22 to 0.11 mL/min/1.73 m^2; $P = 0.075$).

DISCUSSION

MAIN FINDINGS. ERC was effective and safe in patients with and without CKD in this prespecified secondary analysis of EAST-AFNET 4. ERC did not adversely affect kidney function over the follow-up period of 2 years. Without ERC, patients with CKD are more likely to develop recurrent AF. Based on these hypothesis-generating findings, ERC therapy should be considered in the treatment plans of patients with AF and CKD (also see Central Illustration).

ERC REDUCES CARDIOVASCULAR EVENTS ACROSS THE SPECTRUM OF KIDNEY DISEASES. ERC retained its effectiveness across the spectrum of kidney disease severities in this analysis without safety concerns. This is consistent with 2 recent population-based cohort studies comparing (early) rhythm control to rate control, confirming the safety of ERC in patients with CKD in a broader population.[27,28] ERC led to similar rates of sinus rhythm at 2 years. CKD increased AF recurrences with UC, potentially illustrating a higher need for rhythm control therapy in patients with CKD (Table 3) considering that sinus rhythm and AF burden reduction mediate the outcome-reducing effect of ERC.[20,29] More analyses on AF burden in patients with AF and CKD are warranted.

AF ablation is recommended as first-line rhythm control therapy in patients with paroxysmal AF.[30-32] A recent analysis from a prospective multicenter patient cohort found generally favorable outcomes across the GFR spectrum.[33] Because procedure times are getting shorter with more experience and modern ablation techniques,[34,35] a possible benefit for catheter ablation could be expected in patients with cardiovascular and renal comorbidities.[36] Whether AF ablation is effective and safe in patients with multiple comorbidities, including patients with CKD, can in the future be deduced from the outcomes of ongoing trials including $\mathrm{EAST}^{\mathrm{high}}$-AFNET 11 (Early Atrial Fibrillation Ablation for Stroke Prevention in Patients With High Comorbidity Burden; NCT06324188), CABA-HFPEF-DZHK27 (CAtheter-Based Ablation of atrial fibrillation compared to conventional treatment in patients with Heart Failure with Preserved Ejection Fraction),[37] and EAST STROKE (Early Treatment of Atrial Fibrillation for Stroke Prevention Trial in Acute STROKE; NCT05293080).[7]

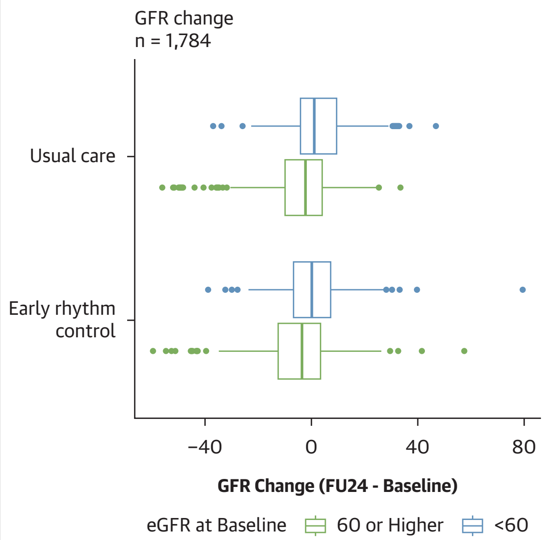
TABLE 6 Change in Estimated GFR at 2-Year Follow-Up by Randomized Group and by Presence of Absence of Kidney Disease

	GFR ≥60 mL/min/1.73 m^2		GFR <60 mL/min/1.73 m^2	
	ERC (n = 1,078)	UC (n = 1,044)	ERC (n = 290)	UC (n = 330)
Change in GFR from baseline to 2 y of follow-up	−3 (−13 to 3)	−2 (−10 to 4)	0 (−7 to 7)	1 (−4 to 10)
Unknown	364	348	112	134
GFR at baseline	79 (70-89)	79 (70-89)	50 (45-56)	52 (44-56)
GFR at 2 years of follow-up	75 (64-87)	77 (67-87)	50 (41-59)	52 (40-61)
Unknown	364	348	112	134

Values are median (Q1-Q3) or n.
Abbreviations are as in Tables 1 and 2.

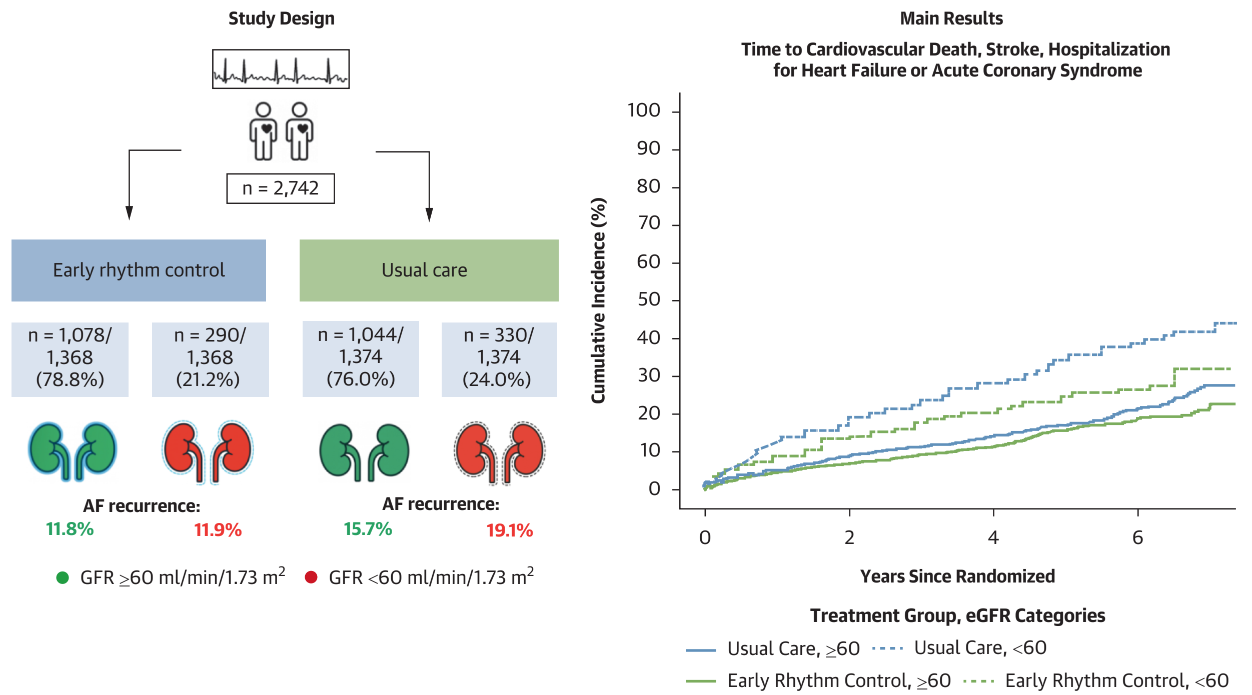
SAFETY OF ERC IN PATIENTS WITH CKD. ERC appeared safe across the spectrum of CKD stages studied (CKD category I-IV), in line with 2 recent reports identifying no safety signals with rhythm control strategies with and without presence of mild or moderate CKD.[27,28] Adverse events related to antiarrhythmic drug therapy were overall infrequent and evenly distributed in GFR groups. No relevant interaction was observed between treatment assignment and kidney function, suggesting that ERC does not disproportionately increase the drug-related complication risk in CKD patients. These findings challenge contemporary concerns about ERC

FIGURE 5 Box Plot Displaying Change in GFR From Baseline to the FU24



eGFR and GFR measured in mL/min/1.73 m^2. Abbreviations as in Figures 1 and 4.

CENTRAL ILLUSTRATION Effectiveness and Safety of Early Rhythm Control in Patients With Atrial Fibrillation and Chronic Kidney Disease



I Early rhythm control therapy is effective and safe across all kidney function stages in the EAST-AFNET 4 trial.
II Recurrent AF appears more common in patients with chronic kidney disease with a potentially greater treatment effect of early rhythm control.
III Early rhythm control therapy does not affect change in kidney function two years after therapy initiation.
IV Chronic kidney disease increases risk of cardiovascular events in patients with recently diagnosed AF.

Schenker N, et al. JACC. 2026;88(2):227-241.

Study design and main results of the prespecified subanalysis of the EAST-AFNET 4 (Early Treatment of Atrial Fibrillation for Stroke Prevention Trial 4) trial.

therapy, eg, reflected in a recent European Heart Rhythm Association survey where most centers limited rhythm control therapy to patients with mild CKD or normal kidney function.[18] Results from ORBIT-AF, a U.S.-based registry, confirm a lower use of rhythm control therapy in patients with advanced kidney disease.[38] The underlying reason for these treatment patterns might be concerns about accumulation of antiarrhythmic drugs: Sotalol can easily accumulate in patients with kidney disease and dronedarone cannot be used with GFRs <30 mL/min/1.73 m^2.[39] Flecainide appeared safe in patients with CKD in this analysis, in line with a recent analysis of EAST-AFNET 4 and a statement from the European Heart Rhythm Association.[39,40] These results suggest that flecainide can be used in selected patients with heart failure with preserved ejection fraction and/or revascularized coronary artery disease[39,40] and in patients with mild or moderate CKD (this analysis). AF ablation appeared safe with and without CKD in this analysis. Major bleeding events were mainly observed in patients with CKD, probably due to the increased bleeding risk in this patient cohort.[41] The use of AF ablation in EAST-AFNET 4 was lower than in contemporary cohorts, reflecting therapy access and recommendations at the time.[24] These findings may help to select rhythm control therapies in patients with AF and kidney disease.

Importantly, ERC did not accelerate kidney function decline in EAST-AFNET 4 patients (**Figure 5**, Supplemental Figure 1), illustrating the safety of contemporary ERC for kidney function.

EFFECT OF ERC ON AF BURDEN AND SECONDARY OUTCOMES IN PATIENTS WITH CKD. Recurrent AF was more common in patients with CKD randomized to UC. ERC therapy appeared more effective in maintaining sinus rhythm with CKD

($P_{\text{interaction}} = 0.036$), and AF burden on ERC was not affected by kidney disease, illustrating the effectiveness of ERC independent of kidney function. This result encourages the use of ERC therapy in patients with CKD. Other secondary outcomes were comparable between kidney function groups, consistent with the main trial.[3]

EFFECT OF KIDNEY DISEASE ON CARDIOVASCULAR OUTCOMES. CKD increases the risk of cardiovascular events in patients with cardiovascular diseases and in the general population.[42] This subanalysis of EAST-AFNET 4 confirms a clear effect of CKD on cardiovascular events over a 5-year timeframe with a near-doubling of risk of cardiovascular events, driven by worsening heart failure and cardiovascular death. Mutual aggravation of CKD, AF, and comorbidities can explain the higher number of heart failure hospitalizations in patients with CKD in this trial (Table 2). Patients with recently diagnosed AF and reduced kidney function also spent more nights in hospital than patients without CKD, similar to unselected patients with CKD.[43,44] Similar to the overall trial, ERC did not reduce hospital nights in patients with CKD, possibly due to hospitalizations not related to AF.[43,44] Estimated GFR has been proposed as a refinement for stroke risk prediction in patients with AF.[45] Our findings support this, as CKD was associated with a nearly 2-fold increased stroke rate in EAST-AFNET 4 (Table 2).

STUDY LIMITATIONS. This study has several limitations. First, although prespecified, this analysis is secondary, contains post-hoc elements, and should therefore be regarded as hypothesis-generating. Three quarters of patients had no CKD, limiting the power, especially in patients with severe kidney disease. Second, kidney outcomes such as acute kidney injury and requirement for dialysis were not captured systematically. Major adverse kidney events composites can therefore not be reported. Third, kidney function was only available at 2 years of follow-up and that information was missing in approximately one third (958 of 2,742) of patients. Small effects of ERC on kidney function may require more patients or longer observation times to become apparent. Fourth, etiology of kidney disease was not available in this analysis. Fifth, long-standing AF may have effects on kidney function not detected in this cohort of patients with a history of AF of 12 months or less. Sixth, EAST-AFNET 4 did not enroll patients with end-stage kidney disease or those on dialysis.[17] Further studies are warranted to determine the safety and effectiveness of ERC in this high-risk population. Seventh, as in any subgroup analysis, selection bias and residual confounding cannot be fully excluded, despite adjustment for multiple covariates.

CONCLUSIONS

In this prespecified analysis of the EAST-AFNET 4 trial, ERC effectively and safely reduced a composite outcome of death, stroke, hospitalization for heart failure, and acute coronary syndrome with and without CKD. Without ERC, conversely, patients with CKD are at increased risk of recurrent AF. These findings support the use of ERC in patients with AF and CKD, while highlighting the need for further dedicated studies in this field.

DATA AVAILABILITY Access to all data will be made available on request (info@kompetenznetz-vorhofflimmern.de).

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PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: ERC is the emerging default treatment choice in patients with recently diagnosed AF. Based on this secondary analysis of the EAST-AFNET 4 trial, the effectiveness and safety of ERC is maintained with and without CKD.

COMPETENCY IN PATIENT CARE: Patients and care teams should include the beneficial effects of ERC when devising shared therapy plans for patients with AF and CKD.

TRANSLATIONAL OUTLOOK: Rhythm control appears effective across the spectrum of no to moderate kidney disease, without effects on kidney function after 2 years. Prespecified analyses of ongoing clinical trials help to further define the effectiveness and safety of rhythm control in patients with AF and kidney disease. Dedicated trials may be needed in patients with AF and advanced kidney disease requiring hemodialysis. Mechanistic research into the interactions between kidneys and atria is warranted to advance this emerging field.

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APPENDIX For supplemental figures and tables, please see the online version of this paper.